SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Technology Stack

Date	04 July 2026
Team ID	SWTID-2026-9325
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	*React.js + Vite*	*React provides excellent component reusability and dynamic state management, which are vital for building a highly responsive, low-stress user dashboard. Vite acts as a modern, ultra-fast build tool that offers hot module reloading (HMR) to speed up frontend development and optimize production builds.*

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	*Python + FastAPI*	*Python offers powerful libraries for financial data processing. FastAPI was selected because it is a high-performance web framework designed for building rapid, robust REST APIs with low latency. It automatically handles async operations and data validation, ensuring lightning-fast communication between the client and backend.*
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	*SQLite + SQLAlchemy ORM*	*SQLite provides a lightweight, serverless relational database that securely handles user credentials, loan structures, and historical records without heavy configuration infrastructure. SQLAlchemy ORM is utilized for strong relational mapping, allowing the team to write clean, secure database operations while keeping the architecture flexible for future cloud migration.*
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	*Python Virtual Environment (venv)*	*Used to isolate application dependencies, maintain package version control, and prevent conflicts across different development machines. It keeps the development environment completely portable, ensuring the system can be cloned and deployed instantly on standard hardware configurations.*

5	**Version Control & CI/CD**	*Git + GitHub*	*Git provides robust source code management and localized version control history. GitHub is used for cloud-based repository hosting to facilitate seamless team collaboration, branch tracking, code reviews, and centralized project management throughout the development lifecycle.*
	(e.g., GitHub, GitLab, Jenkins)		
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	*Google Gemini API + JWT (JSON Web Tokens)*	*The Google Gemini API is the core AI service that powers intelligent debt-stress analysis, settlement predictions, and the professional generation of lender negotiation request letters. JWT enables secure, token-based user authentication and protected session management across backend endpoints.*